Zongzhe Xu

Los Angeles, CA
zongzhex@ucla.edu | (+1) (314) 319-2028 | zongzhe-xu.github.io/zongzhe_homepage

EDUCATION

University of California, Los Angeles 2025 – Present
Ph.D. Student in Computer Science

- Amazon Fellow, 2026

Carnegie Mellon University, PA 2023 – 2024
Master of Science in Machine Learning

Washington University in St. Louis, MO 2019 – 2023
B.S. in Computer Science and Mathematics | Minor in Bioinformatics | Minor in Biology *Summa Cum Laude*

- Major GPA: 3.99/4.0 · Dean's List for every semester · Dean Select Fellowship - $5,000 Annually

RESEARCH INTERESTS

AI for health, foundation models, multimodal learning, time series

SELECTED PUBLICATIONS

** denotes equal contribution*

[1] **Inertia-1: An Open Exploration of Wearable Motion Foundation Models**
Z. Xu*, A. Anand*, S. Jiang*, C. Zhuang, Z. Shuai, S. Sankararaman, and Y. Yang
Conference on Neural Information Processing Systems (NeurIPS), 2026
Oral, NeurIPS 2026 GenAI4Health Workshop

[2] **SleepLM: Natural-Language Intelligence for Human Sleep**
Z. Xu, Z. Shuai, E. Mozaffari, R. S. Aysola, R. Kumar, and Y. Yang
International Conference on Machine Learning (ICML), 2026 · **Spotlight**

[3] **OSF: On Pre-training and Scaling of Sleep Foundation Models**
Z. Shuai, **Z. Xu**, D. Yang, W. Wang, and Y. Yang
International Conference on Machine Learning (ICML), 2026

[4] **Signal or Noise? Understanding Generative Models for Real-World Sensor Time Series**
Z. Shuai*, **Z. Xu***, Y. Wu*, S. Li, T. Li, and Y. Yang
arXiv preprint, 2026

[5] **This Time is Different: An Observability Perspective on Time Series Foundation Models**
B. Cohen, E. Khwaja, … **Z. Xu**, … D. Asker, A. Talwalkar, O. Abou-Amal
Conference on Neural Information Processing Systems (NeurIPS), 2025

[6] **Specialized Foundation Models Struggle to Beat Supervised Baseline**
Z. Xu*, R. Gupta*, W. Cheng, A. Shen, J. Shen, A. Talwalkar, and M. Khodak
International Conference on Learning Representations (ICLR), 2025

[7] **V2X-DG: Domain Generalization for Vehicle-to-Everything Cooperative Perception**
B. Li*, **Z. Xu***, J. Li, X. Liu, J. Fang, X. Li, and H. Yu
IEEE International Conference on Robotics and Automation (ICRA), 2025

RESEARCH EXPERIENCE

HAIL Lab, University of California, Los Angeles 2025 – Present
Graduate Student Researcher | Supervised by Dr. Yuzhe Yang, UCLA

- Develop foundation models that learn transferable health representations from heterogeneous, multimodal, and irregular time-series data such as sleep and wearable biosignals.
- Build language interfaces that let models describe, retrieve, and reason about raw physiological signals.

Datadog, Inc. 2025
Data Scientist Intern

- Surveyed modern time series foundation models and developed a unified evaluation framework to benchmark 14 state-of-the-art agentic, pretrained, and supervised forecasting models under standardized preprocessing, inference, and evaluation protocols.

- Conducted large-scale empirical evaluation of zero-shot and supervised forecasting models on observability and public time series benchmarks, enabling fair and reproducible comparison across model families.
- Contributed to TOTO, a forecasting foundation model for observability data, by proposing timestamp-aware positional embeddings that encode real temporal spacing and granularity to better capture frequency-dependent structure in time series.

Sage Lab, Carnegie Mellon University 2023 – 2024
Graduate Student Researcher | Supervised by Dr. Ameet Talwalkar, Carnegie Mellon University

- Critically assessed if the significant resources required to train and deploy foundation models in three specialized domains (genomics, time series, and satellite imaging) are always justified by a substantial performance gain. Discovered that traditional supervised methods could achieve competitive, if not superior, performance when appropriately configured and optimized.
- Enhanced cross-modal fine-tuning to adapt LLMs for scientific tasks, integrating regularization strategies, and refining alignment approaches for improved robustness across modalities.

Cleveland Vision & AI Lab, Cleveland State University 2024
Research Assistant | Supervised by Dr. Hongkai Yu, Cleveland State University

- Deviced algorithms with Mix-up representations of point clouds from ego&nonego vehicles and alignment methods that increases the generalizability of the algorithm across diverse test time sim&real environments
- Implemented and tested a novel adversarial algorithm for 3D multiagent perception. The adversarial algorithm is robust to multiple potential communications corrupts during cooperations.

DAMO academy, Alibaba | Sichuan Digital, Shudao Investment Group 2023
Research Intern

- Built a training-free few-shot classifier on top of existing models to improve open-vocabulary object detection for traffic monitoring under abnormal conditions (tunnels, rainy weather, traffic jams).

OBGYN Department, Washington University School of Medicine 2021 – 2023
Part-time Clinical Researcher | Supervised by Dr. Peinan Zhao, Washington University in St Louis

- Contributed end to end to the longitudinal research from sensor data collection & calibration to ultrasounds & pressure analysis to inspect the relationship between cervical softening and pre-term birth.
- Designed edge detection & track system to track cervical compression in ultrasound images to calculate the cervical young's modulus. Integrated various ML methods throughout the project (Laplacian filter & Fourier decomposition for image denoising, group linear regression over patient records for trend detection)
- Proposed and constructed a hardware + software system that synchronizes the data input from sensors and loadcells and provide real time visualizations to simplify the operations in real clinical tests. Automated the processing program to accurately capture cervical structure in real time.

TEACHING EXPERIENCE

James McKelvey School of Engineering, Washington University in St. Louis 2021 – 2022
Teaching Assistant

- CSE 412 Introduction to Artificial Intelligence, CSE 417 Machine Learning Theory; held 4-6 weekly office hours and assisted with grading for classes of 70+ students.

SERVICE

Conference Reviewer: ICLR 2025, 2026 · ICML 2026 · NeurIPS 2025, 2026 · KDD 2025 · ICASSP 2025

SKILLS

Languages: Python, R, Java, C++, SQL, C | **Frameworks & Tools:** PyTorch, Keras, MCP, vLLM